

- 174.** The value of $5^2 + 6^2 + \dots + 10^2 + 20^2$ is
 (a) 755 (b) 760
 (c) 765 (d) 770
- 175.** Given that $1 + 2 + 3 + 4 + \dots + 10 = 55$, then the sum $6 + 12 + 18 + 24 + \dots + 60$ is equal to
 (a) 300 (b) 330
 (c) 455 (d) 655
- 176.** If m and n are natural numbers such that $2^m - 2^n = 960$, what is the value of m ? (M.A.T., 2007)
 (a) 10 (b) 12
 (c) 15 (d) Cannot be determined
- 177.** On multiplying a number by 7, all the digits in the product appear as 3's. The smallest such number is (C.P.O., 2006)
 (a) 47619 (b) 46719
 (c) 48619 (d) 47649
- 178.** The number of digits in the smallest number, which when multiplied by 7 yields all nines, is
 (a) 3 (b) 4
 (c) 5 (d) 6
- 179.** A boy multiplies 987 by a certain number and obtains 559981 as his answer. If in the answer both 9's are wrong but the other digits are correct, then the correct answer will be
 (a) 553681 (b) 555181
 (c) 555681 (d) 556581
- 180.** The numbers 1, 3, 5,, 25 are multiplied together. The number of zeros at the right end of the product is (R.R.B., 2006)
 (a) 0 (b) 1
 (c) 2 (d) 3
- 181.** The numbers 1, 2, 3, 4,, 1000 are multiplied together. The number of zeros at the end (on the right) of the product must be
 (a) 30 (b) 200
 (c) 211 (d) 249
- 182.** First 100 multiples of 10 i.e. 10, 20, 30,, 1000 are multiplied together. The number of zeros at the end of the product will be
 (a) 100 (b) 111
 (c) 124 (d) 125
- 183.** The number of zeros at the end of the product $5 \times 10 \times 15 \times 20 \times 25 \times 30 \times 35 \times 40 \times 45 \times 50$ is
 (a) 5 (b) 7
 (c) 8 (d) 10
- 184.** The number of zeros at the end of $60!$ is
 (a) 12 (b) 14
 (c) 16 (d) 18
- 185.** The numbers 1, 3, 5, 7,, 99 and 128 are multiplied together. The number of zeros at the end of the product must be
 (a) Nil (b) 7
 (c) 19 (d) 22
- 186.** The numbers 2, 4, 6, 8,, 98, 100 are multiplied together. The number of zeros at the end of the product must be
 (a) 10 (b) 11
 (c) 12 (d) 13
- 187.** Let S be the set of prime numbers greater than or equal to 2 and less than 100. Multiply all the elements of S . With how many consecutive zeros will the product end?
 (a) 1 (b) 4
 (c) 5 (d) 10
- 188.** Find the number of zeros at the end of the result $3 \times 6 \times 9 \times 12 \times 15 \times \dots \times 99 \times 102$.
 (a) 4 (b) 6
 (c) 7 (d) 10
- 189.** The unit's digit of 13^{2003} is (A.A.O. Exam, 2010)
 (a) 1 (b) 3
 (c) 7 (d) 9
- 190.** The digit in the unit's place of the number 123^{99} is (I.A.M., 2007)
 (a) 1 (b) 4
 (c) 7 (d) 8
- 191.** Match List I with List II and select the correct answer:
- | List I
(Product) | List II
(Digit in the unit's place) |
|---------------------|--|
| A. $(1827)^{16}$ | (1) 1 |
| B. $(2153)^{19}$ | (2) 3 |
| C. $(5129)^{21}$ | (3) 5 |
| | (4) 7 |
| | (5) 9 |
| A B C | A B C |
| (a) 1 4 3 | (b) 4 2 3 |
| A B C | A B C |
| (c) 1 4 5 | (d) 4 2 5 |
- 192.** The digit in the unit's place of the number $(67)^{25} - 1$ must be
 (a) 0 (b) 6
 (c) 8 (d) None of these
- 193.** The unit's digit in the product $274 \times 318 \times 577 \times 313$ is
 (a) 2 (b) 3
 (c) 4 (d) 5
- 194.** In the product $459 \times 46 \times 28^* \times 484$, the digit in the unit place is 8. The digit to come in place of $*$ is
 (a) 3 (b) 5
 (c) 7 (d) None of these
- 195.** The digit in the unit place of the number represented by $(7^{95} - 3^{58})$ is
 (a) 0 (b) 4
 (c) 6 (d) 7
- 196.** Unit's digit in $(784)^{126} + (784)^{127}$ is
 (a) 0 (b) 4
 (c) 6 (d) 8